



5369 - A Database of Interstellar Ice Absorption Spectra

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Volker Tolls (PI)	Smithsonian Institution Astrophysical Observatory
Dr. Matthew L. N. Ashby (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Joseph L. Hora (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Howard A. Smith (CoI)	Smithsonian Institution Astrophysical Observatory

OBSERVATIONS

ABSTRACT

Throughout star formation ices play a very important role. They grow to be a major reservoir for the volatile elements H, O, C, N, P, and S (CHNOPS). The chemical composition and its evolution from molecular clouds to protoplanetary disks (PPD) eventually determines the atomic and molecular composition of forming planets, which coagulated from dust and pebbles in the PPD. Many chemical reactions take place on dust grains in the coldest parts ($T < 30\text{K}$) of molecular clouds, cloud cores, and the midplane of PPDs where they are shielded from the destructive UV radiation of (proto)stars. These chemical reactions produce not only molecules like H_2O , but also more complex molecules like CH_3OH , hydrocarbons, alcohols, and aldehydes. To quantify the chemical reaction efficiencies and how they affect the chemical composition, inherited or reprocessed, we must determine the bulk composition from molecular clouds to PPDs. This archival research proposal aims at providing the first step to quantify the ice composition during the various steps of star formation by accumulating all ice absorption spectra from JWST observations, currently >140 spectra. We will use the database to categorize the observations, to investigate the correlations between ice species and types of observed absorbing media, and to look for evolutionary trends, extending previous efforts (Öberg et al. 2011) based on only a limited number of ISO, AKARI, and ground-based ice observations. The new database of ice observations is just the first step, and over time, JWST will observe many more astronomical objects and the database will grow. Finally, the database will be made publicly available through the MAST archive.

OBSERVING DESCRIPTION